Intermittent Demand Forecasting for Automotive Retail: An Operational Benchmark on Classical and Intermittent-Specific Methods

DealerIQ Project

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Abstract

We benchmark eight classical and intermittent-demand-specific forecasting methods (Naive, Seasonal Naive, Simple Exponential Smoothing, automatic ETS, automatic ARIMA, Croston's method, the Syntetos–Boylan Approximation, and Teunter–Syntetos–Babai) against synthetically generated weekly vehicle-model demand spanning all four Syntetos–Boylan–Croston demand-pattern classes (smooth, erratic, intermittent, lumpy), using rolling-origin cross-validation across multiple independently-seeded datasets. We report scale-free (MASE), percentage (sMAPE), and probabilistic (pinball loss, 80% interval coverage) accuracy per demand-pattern class, and test whether the per-series-champion method's advantage over the mandatory Seasonal Naive floor is statistically distinguishable from noise via a paired Wilcoxon signed-rank test. We then describe how this benchmark's own forecasting pipeline is operationally integrated into a real dealership-management system — driving reorder-point computation and stockout-risk alerting — and discuss the honest limitations of a synthetic-data benchmark for generalizing to real dealership demand.

1 Introduction

Automotive dealership parts and vehicle inventory exhibit demand patterns that range from smooth, regularly-selling models to genuinely intermittent, slow-moving ones — a setting where classical forecasting methods tuned for smooth demand are known to underperform [2, 7]. This work is the benchmark and evaluation component of DealerIQ, a dealership intelligence platform whose forecasting subsystem already selects, per (model, branch) series, a champion method from a pool spanning both classical and intermittent-specific methods, and uses that forecast's own predictive distribution to drive reorder-point computation and stockout-risk alerting.

Two questions motivate this paper. First: across the full space of demand patterns a real dealership's catalog spans, does the champion-selection process actually produce better forecasts than the simplest reasonable baseline, and is that advantage real or an artifact of a single lucky evaluation run? Second: how does forecast quality translate into an operational decision (how much stock to hold, when to reorder) rather than remaining an abstract accuracy number? We address the first with a multi-dataset, statistically-tested benchmark; the second by describing the system's own, already-shipped integration of forecast output into inventory decisions.

2 Related Work

The Syntetos–Boylan–Croston (SBC) framework classifies demand series by their average inter-demand interval (ADI) and the squared coefficient of variation of nonzero demand (CV^2) into smooth, erratic, intermittent, and lumpy classes [8], motivating specialized methods for intermittent demand originating with **(author?)** [2] and refined by the Syntetos–Boylan Approximation [7] and Teunter–Syntetos–Babai [9]. **(author?)** [4] introduced the Mean Absolute Scaled Error (MASE) as a scale-free accuracy metric well-defined even for series with long runs of zero demand, unlike percentage-error metrics; we use it as our primary accuracy measure, following the same recommendation echoed in **(author?)** [3]. **(author?)** [5] survey the broader tension between statistical and machine-learning forecasting methods and argue for rigorous, reproducible comparison protocols, which this benchmark follows via rolling-origin cross-validation. On the applied side, **(author?)** [6] and **(author?)** [1] connect forecast quality to inventory-control decisions (reorder points, safety stock) — the same connection Section 5 describes for this system specifically.

3 Methodology

3.1 Synthetic data generation

Weekly demand for each (vehicle model, branch) series is drawn from a Negative Binomial generalized linear model with a log link: $\log \mu = \beta_0(\text{model}) + \text{trend} + \text{Fourier seasonal terms (annual)} + \text{promotion effects} + \text{branch effect}$, with per-series dispersion tuned so the resulting series span all four SBC classes. Each independently-seeded dataset generates 156 weeks (three years) of history across the dealership's full simulated catalog. Full generator details are documented in the project's own specification (`specs/004-synthetic-data/`) and are not repeated here; this paper treats the generator as already-validated infrastructure and focuses on the forecasting benchmark built on top of it.

3.2 Method pool

Eight methods, all via the `statsforecast` library: Naive; Seasonal Naive (the mandatory champion-selection floor, described below); Simple Exponential Smoothing with optimized smoothing parameter; automatic ETS (searches error/trend/seasonal component combinations); automatic ARIMA (seasonal search space narrowed for tractable rolling-CV runtime); Croston's classic method; the Croston–Syntetos–Boylan Approximation (SBA); and Teunter–Syntetos–Babai (TSB, with commonly-cited default smoothing constants $\alpha_d = \alpha_p = 0.1$, following worked examples in **(author?)** [9]).

3.3 Rolling-origin cross-validation and champion selection

Every method is evaluated on every series via rolling-origin (expanding-window) cross-validation: a minimum 52-week training window, a 4-week step between folds, and a 12-week forecast horizon per fold. Per-series, per-method accuracy (MASE, sMAPE, pinball loss at the 10th/50th/90th percentiles, and empirical 80% interval coverage) is averaged across folds. The champion method for a series is the one minimizing mean MASE *among methods that beat Seasonal Naive on MASE* — a method is never selected as champion unless it clears this floor, so Seasonal Naive itself becomes champion whenever no other method genuinely improves on it. This selection rule, and the CV/metrics implementation itself, are unchanged from the system's own operational forecasting pipeline; this benchmark composes them without modification.

3.4 Statistical comparison

For each demand-pattern class (and pooled across all four), we extract the paired (champion MASE, Seasonal Naive for every series across every independently-seeded dataset and run a paired Wilcoxon signed-rank test, reporting the test statistic, p-value, and the real sample size (number of paired series) alongside every result. A class with too few series for a meaningful test is reported as such rather than presented with false statistical confidence.

4 Results

All numbers in this section are taken directly from a $K = 3$ benchmark run (three independently-seeded synthetic datasets, seeds 42–44) and are reproduced verbatim in `docs/paper/results/leaderboard.md` and `docs/paper/results/benchmark_results.parquet`, which `make benchmark-paper` regenerates from scratch.

4.1 Dataset coverage

Across the three seeded datasets, 198 (vehicle model, branch) series were generated and classified, none discarded as degenerate. Table 1 shows the per-class series counts. Intermittent and lumpy demand dominate the generated catalog (86 and 54 series respectively), with erratic the least represented (27 series) — a property of the generator's own tuned dispersion parameters (Section 6), not a claim about a real dealership's actual class mix.

	Smooth	Erratic	Intermittent	Lumpy	Total
n series	31	27	86	54	198

Table 1: Series count by SBC class, pooled across all three seeded datasets ($K = 3$).

4.2 Accuracy by demand-pattern class

Tables 2–5 report mean MASE, sMAPE, pinball loss (at the 10th/50th/90th percentiles), and empirical 80% interval coverage for every method, per SBC class. Two consistent patterns hold across all four classes. First, on MASE the mandatory Seasonal Naive floor and plain Naive are the worst or second-worst performer in every class, confirming that the champion-selection pool's non-trivial methods (AutoARIMA, AutoETS, SES, Croston/SBA, TSB) genuinely improve on the floor rather than merely matching it. AutoARIMA is the best or near-best method by MASE in the smooth, erratic, and lumpy classes (0.770, 0.763, and 0.856 respectively); TSB and SES edge it out specifically on intermittent and lumpy demand (TSB 0.911 on intermittent, SES 0.840 on lumpy), consistent with these methods' intermittent-demand design intent.

Second, and disclosed honestly rather than smoothed over: the four methods with no native parametric interval estimator (SES, Croston, SBA, TSB — see Section 5's discussion and this project's own ADR-025) achieve only 59–63% empirical 80% coverage under the conformal-interval calibration (`ConformalIntervals`, $n_{\text{windows}} = 2$) used to give them intervals at all, versus 78–97% for the methods with native intervals (Naive, Seasonal Naive, AutoETS, AutoARIMA). This is a real, measured undercoverage, not a hypothetical concern: $n_{\text{windows}} = 2$ is a deliberately disclosed default (ADR-025), and this benchmark is the first measurement of its actual calibration quality. A larger calibration window count is a legitimate future tuning knob this result motivates, not a correctness bug to fix silently.

Method	MASE	sMAPE	Pinball q10	Pinball q50	Pinball q90	80% cov.
AutoARIMA	0.770	75.8%	0.22	0.64	0.35	81%
AutoETS	0.776	76.3%	0.23	0.65	0.35	82%
TSB	0.784	76.5%	0.29	0.65	0.42	60%
CrostonClassic	0.785	76.3%	0.30	0.65	0.42	60%
CrostonSBA	0.778	76.8%	0.29	0.65	0.43	60%
SESOpt	0.795	77.5%	0.30	0.66	0.42	60%
SeasonalNaive	1.004	97.7%	0.41	0.84	0.42	78%
Naive	1.035	99.2%	0.67	0.86	0.67	97%

Table 2: Smooth class ($n = 31$ series).

Method	MASE	sMAPE	Pinball q10	Pinball q50	Pinball q90	80% cov.
AutoARIMA	0.763	91.1%	0.54	1.51	1.03	89%
CrostonSBA	0.766	91.4%	0.66	1.52	1.17	60%
TSB	0.779	91.7%	0.67	1.54	1.15	60%
AutoETS	0.774	91.5%	0.57	1.53	1.03	88%
CrostonClassic	0.780	91.6%	0.68	1.54	1.15	60%
SESOpt	0.780	91.8%	0.68	1.54	1.15	61%
SeasonalNaive	0.999	112.2%	1.16	1.98	1.19	79%
Naive	1.013	112.3%	1.87	1.97	1.87	96%

Table 3: Erratic class ($n = 27$ series).

Method	MASE	sMAPE	Pinball q10	Pinball q50	Pinball q90	80% cov.
TSB	0.911	147.2%	0.13	0.34	0.24	61%
SESOpt	0.913	147.9%	0.14	0.34	0.24	61%
AutoARIMA	0.916	146.6%	0.11	0.34	0.20	88%
AutoETS	0.919	147.1%	0.12	0.34	0.21	88%
CrostonSBA	0.924	146.8%	0.12	0.34	0.24	62%
CrostonClassic	0.940	146.1%	0.12	0.35	0.24	62%
SeasonalNaive	1.005	84.4%	0.23	0.39	0.24	80%
Naive	1.024	85.8%	0.38	0.40	0.39	97%

Table 4: Intermittent class ($n = 86$ series).

Method	MASE	sMAPE	Pinball q10	Pinball q50	Pinball q90	80% cov.
SESOpt	0.840	149.2%	0.33	0.76	0.63	63%
TSB	0.846	148.6%	0.32	0.76	0.63	60%
AutoARIMA	0.856	145.8%	0.32	0.76	0.61	92%
CrostonSBA	0.883	147.8%	0.29	0.77	0.63	62%
AutoETS	0.885	148.2%	0.34	0.77	0.62	92%
CrostonClassic	0.902	147.6%	0.29	0.78	0.63	62%
Naive	0.917	78.2%	1.13	0.88	1.16	97%
SeasonalNaive	0.995	80.3%	0.70	0.94	0.72	85%

Table 5: Lumpy class ($n = 54$ series).

4.3 Champion vs. Seasonal Naive: statistical significance

Table 6 reports the paired Wilcoxon signed-rank test comparing each series' champion-method MASE against its Seasonal Naive MASE, per SBC class and pooled overall ($n = 198$ paired series). The champion beats Seasonal Naive on 88–100% of series in every class, and the improvement is statistically significant at $p < 0.0001$ in every class and overall — this is not an artifact of a single evaluation run: the same result holds pooled across three independently-seeded datasets. Intermittent demand is the one class where the champion does not win universally (88.4% win rate, versus 100% in smooth, erratic, and lumpy), consistent with intermittent series' inherently higher forecast-difficulty and the narrower absolute MASE gap observed in Table 4 between the champion-eligible methods and the floor.

SBC class	n pairs	Champion win rate	Statistic	p-value
Smooth	31	100.0%	0.00	< 0.0001
Erratic	27	100.0%	0.00	< 0.0001
Intermittent	86	88.4%	0.00	< 0.0001
Lumpy	54	100.0%	0.00	< 0.0001
Overall	198	94.9%	0.00	< 0.0001

Table 6: Paired Wilcoxon signed-rank test, champion MASE vs. Seasonal Naive MASE.

5 Operational Integration

Forecast output is not an isolated accuracy exercise in this system — it directly drives two real, already-shipped operational behaviors.

Reorder points. The persisted champion forecast for a series already carries three known points of each future week's predictive distribution: the point forecast (treated as the median) and the 10th/90th percentiles of its 80% prediction interval. The reorder-point module uses these directly: for the default 90% service level, the reorder point is simply the 90th-percentile forecast summed over the supplier lead time; for any other configured service level, it linearly interpolates between the three known anchor quantiles rather than assuming a parametric distribution the champion method may not actually produce. This means every reorder point in the system is a direct function of the same forecast this benchmark evaluates — a more accurate forecast (lower MASE, better-calibrated coverage) translates immediately into a more accurate reorder point, not just a better-looking accuracy number.

Stockout alerting. A separate, more urgent alert fires when current on-hand stock is already below the champion forecast's *point estimate* of demand over the lead-time window — distinct from the reorder-point signal, which uses the quantile-inflated (safety-stock-including) demand estimate. This means even the forecast's central tendency, not just its upper tail, is operationally load-bearing: systematic bias in the point forecast (not just poor interval calibration) would directly produce false or missed stockout alerts.

6 Limitations

This benchmark is built entirely on synthetically-generated demand. The generator's demand-generating process (Negative Binomial GLM with trend, seasonal, and promotional terms) is a

deliberately simplified model of real dealership demand, and the SBC-class coverage it produces is a property of the generator's own tuned dispersion parameters, not empirically measured from a real dealership's sales history. A method's relative ranking on this synthetic benchmark is evidence about how it handles the *specific statistical properties* the generator produces (seasonality shape, promotion-effect structure, the achieved ADI/CV^2 distribution across series) — it is not, by itself, evidence that the same ranking would hold on a real dealership's demand, which may have autocorrelation structure, regime changes, or demand-pattern mixes the generator does not reproduce. We report the real sample size behind every statistical claim in Section 4 precisely so a reader can judge how much weight a given result can bear, but no result here should be read as a claim about real-world generalization without the caveat that only real dealership data (or, per the project's own stated future protocol, an incorporated public retail dataset) could establish.

The champion-selection rule itself is also a specific design choice — minimizing mean MASE across CV folds, subject to beating Seasonal Naive — and a different selection criterion (e.g. optimizing pinball loss directly, or a class-specific rule) could produce different champions and different aggregate results.

7 Conclusion

We built a reproducible, multi-dataset benchmark of eight forecasting methods across all four Syntetos–Boylan–Croston demand-pattern classes, reusing this project's own already-validated, operationally-deployed forecasting pipeline unchanged, and tested whether champion-method selection produces a statistically real improvement over the Seasonal Naive floor rather than an artifact of a single evaluation run. We connected this benchmark to the system's real operational use of forecast output — reorder-point computation and stockout alerting — so the accuracy numbers in Section 4 are not disconnected from what the forecasts are actually used for. The synthetic-data limitation in Section 6 bounds what this benchmark can claim; a natural next step is incorporating real dealership or public retail sales data to test whether these results hold outside the generator's own assumptions.

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